ANDREW LUTSKY

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA December 2024
M.S. in Quantitative Biology and Bioinformatics (GPA: 4.12/4.0)
Thesis Topic: *Novel Persistent Embeddings of Protein Pockets and their Evaluation on Proper Train-Test Splits for Ligand Binding Affinity*
Awards: Outstanding Academic Achievement Award | Professional Honors Award | Outstanding Service and Leadership Award | Research Excellence with Honors Award
Coursework included: Programming for Scientists (02601), Essential Math and Statistics (02680), Statistical Computing (03701), Data Analysis (03702), Algorithms and Advanced Data Structures (02613), Machine Learning for Scientists (02620), Bioinformatics Data Practicum (03713), Computational Genomics (03711), Computational Modeling, Statistics, and Machine Learning in Science (09615)

University of Illinois at Urbana-Champaign, Champaign, Illinois May 2023
BS in Biochemistry, Minor in Chemistry; Highest Distinction in Research (GPA: 3.86/4.0)
Thesis Topic: *Brownian Diffusion of Charged Species in Relation to Protein Crowding Effects*
Coursework included: Biochemistry, Biophysics, Computing in Molecular Biology, Endocrinology, Computer Science I, Organic Chemistry I, Organic Chemistry II, Biostatistics, Calculus I – III

University of Stockholm, Stockholm, Sweden January - May 2022
Department of Biochemistry and Biophysics, Traineeship in Neurochemistry I and II
Research Topic: Regulation of in vitro ApoE secretion in hepatoma carcinomic cell lines

RESEARCH EXPERIENCE

August 2023 – March 2025
Koes Research Lab
PI: David Koes; Professor of Computational and Systems Biology at the University of Pittsburgh
Graduate Student Researcher:

- Benchmarked and assessed protein pocket descriptors, to validate and assess a open source protein pocket descriptors. Used these benchmarks to compare against a learned protein pocket descriptor.
- Created a tool as part of a thesis project that utilizes persistence vectors to create a protein pocket descriptor; integrated these descriptors with a GVP-GNN architecture to perform Binding Affinity Regression on PDBbind, PLINDER, and custom datasets to evaluate these topological descriptors on leaky datasets.

July 2022 – August 2023
Theoretical and Computational Biophysics Group
PI: Emad Tajkhorshid; Professor of Biochemistry at University of Illinois at Urbana Champaign
Undergraduate Research Assistant:

- Completed a senior research thesis under the supervision of Professor Tajkhorshid to establish a quantitative relationship of crowding effects on small species in solution, assessing the diffusion of prototypical molecules in the presence of varying proteins utilizing open-sourced software such as namd, charmm-gui, etc.

- Completed a research project on the characterization of lipid raft interactions with anesthetic molecules in flooding simulations in both coarse grained and all-atom simulations using similar experimentation methods from my research thesis.

June 2022 – June 2023
Morrow Labs
PI: Dan Morrow; Professor of Psychology Education, University of Illinois at Urbana-Champaign
Undergraduate Research Assistant
- Assisted Dr. Dan Morrow with critiquing a virtual health agent. The research project aim was to provide feedback on an integrated web health agent while simultaneously evaluating "teach-back" methods for older adults.
- Identified viable participants by coordinating with iHelp, conducted interviews with human participants, and performed data analysis on behavioral tests.

March 2022 - June 2022
Nielsen Lab
PI: Nielsen, Henrietta; Professor of Neurochemistry at Stockholm University
Neurochemistry Trainee
- Cultured HepG2 and Huh7 cell lines, both homozygous for Alzheimer's disease risk-neutral apoE3 under various conditions, including exposure to exogenously added apoE4, to elucidate the mechanisms by which liver-secreted proteins affect neurodegeneration.
- Performed western blotting, ELISA, BCA, and Mammalian Aseptic techniques to characterize protein expression and secretion under various conditions.

June 2021 - December 2021
Illinois Sustainable Technology Center
PI: Rajagopalan, Nandakishore; Illinois Sustainable Technology Center at University of Illinois at Urbana-Champaign
Undergraduate research assistant/chemist intern
- Worked on a project that focused on the degradation of microplastics and dye compounds and their recapture to develop a sustainable way of recapturing dyes and microplastics released into the environment.
- Measured dye degradation under varying conditions of composite plastics. Conducted spectroscopic analysis of plastic composites.

PROFESSIONAL EXPERIENCE

Mar 2026 – Present
Booz Allen Hamilton, Charleston, SC
Machine Learning Engineer
- Developed and supported the delivery of a product that coordinates ratings for thousands of veteran claims relating to burials, using AWS lambda and AWS bedrock.
- Supported and developed a product that leverages LLM's to perform auto highlighting/summarization enabling veteran service representatives to auto annotate veteran claims in seconds instead of hours.
- Collaborated with engineering teams on large-scale data pipelines to clean, structure, and evaluate datasets used to define metrics and requirements for AI-based government contracts.

Aug 2025 – Mar 2026
Medical University of South Carolina, Charleston, SC
Systems Programmer/Developer II

- Modeled CHAMP1 and Hp1a dimer complex utilizing open-source co-folding, course-grained simulation, and all atom simulation software, to determine structural basis for autism related phenotypes.
- Adapted and developed Nextflow pipelines for IMC data processing and analysis, contributing to a workflow that enables for high-throughput processing of Hyperion IMC data.
- Adapted, retrained, and fine-tuned DeepFoci to perform counts and high-throughput confocal image analysis of double-stranded break foci.

Jan 2025 – June 2025
Pharmacological Sciences Department, Icahn School of Medicine at Mount Sinai, New York, NY
Bioinformatics Software Engineer

- Developed and worked on a protocol (*Wiley Current Protocols)* for creating Knowledge Graph User Interfaces in a modular way for web-applications; developed a reasonable network taking advantage of the interfaces modules to do custom graph queries utilizing neo4j.
- Developed parallelization pipelines to process and integrate large perturbation datasets (Tahoe 100M), computed through AWS architectures; Aligned and compared other perturbation datasets like the LINCS L1000 and Ginkgo's GDPx1 and GDPx2 to find overlap in perturbation gene sets and differentially expressed genes.

Aug 2024 – Dec 2024
Department of Computer Science, Carnegie Mellon University
Teaching Assistant, Algorithms and Advanced Data Structures

- Supervised discussion sections, conducted office hours, curated test and quiz questions, and graded papers and homework in the field of Algorithms and Advanced Data Structures taught by Professor Yun (William) Yu. Topics included run-time analysis, divide-and-conquer, dynamic programming, network flow, linear and integer programming, large-scale search algorithms and heuristics, efficient data storage and query, and NP-completeness.
- Held discussion sections for over 50 students, facilitating discussions, delivering short lectures over the course of the semester, and administering quizzes to reinforce topics taught during the main course lectures.

May 2024 – August 2024
Regeneron Pharmaceuticals, Rensselaer, NY
Quality Control/Analytical Sciences Intern

- Developed an RShiny application that allows for the analysis of shotgun proteomics (LC-MS/MS) data (volcano plots, enriched protein families across experimental groups, etc.) and enabling a 100x reduction in reprocessing time compared to commercial software.
- Created a machine learning framework and pipeline to determine optimal product yield from the LC-MS/MS data.

June 2021 – January 2022
Department of Chemistry, University of Illinois at Urbana-Champaign
Organic Chemistry Teaching Assistant

- Selected to be a teaching assistant during my junior year based on outstanding performance in both Organic Chemistry and Organic Chemistry Laboratory.
- Taught a weekly 4-hour chemistry laboratory class and conducted weekly office hours.
- Assisted in the restructuring of the course; provided feedback and professional critique to drive improvements in lab experiments and assignments.

PUBLICATIONS

Evangelista, J. E., **Lutsky, A. D.,** Byrd, A. I., Clarke, D. J. B., Prabhakaran, A., Jenkins, S. L., & Ma'ayan, A. (2025). Creating an Interactive Web Interface for Networks Stored in Knowledge Graph Databases. *Current protocols*, *5*(9), e70200. https://doi.org/10.1002/cpz1.70200

GRANTS/ALLOCATIONS

Andrew Lutsky, Stefano Berto, Medical University of South Carolina; *Modeling and Elucidation of CHAMP1-HP1α Binding Interactions*

THESES AND PRESENTATIONS

Lutsky, Andrew, DeBlasio D., Lugo-Martinez J., Koes, D. *Novel Persistent Embeddings of Protein Pockets and their Evaluation on Proper Train-Test Splits for Ligand Binding Affinity*. Master's defense. 2024, Carnegie Mellon University.
Lutsky, Andrew, Chan, A., Tajkhorshid, E. *Crowding effects on diffusion of small prototypical species in Molecular Dynamics simulations*. Poster presentation at the Undergraduate Research Symposium Spring 2023, University of Illinois at Urbana-Champaign.

AWARDS

- Outstanding Academic Achievement Award (Carnegie Mellon University), 2024
- Professional Honors Award (Carnegie Mellon University), 2024
- Outstanding Service and Leadership Award (Carnegie Mellon University), 2024
- Research Excellence with Honors Award (Carnegie Mellon University), 2024
- Highest Distinction in Research (University of Illinois at Urbana-Champaign), 2023

LEADERSHIP AND ENGAGEMENT

Aug 2023 – Dec 2024
Carnegie Mellon University
Graduate Student Association (GSA) representative for Biological Sciences

- Represented Biological Sciences masters students in the GSA, attending biweekly meetings to discuss and vote on policies affecting students and advocate for funding for graduate student events.

- Planned themed social events to foster community and engagement of a diverse population of graduate students across MS. Computational Biology, MS. Quantitative Biology and Bioinformatics, MS. Biotechnology, MS. Automated Sciences, etc.

2021-2022
University of Illinois at Urbana Champaign
Treasurer, Climbing Club
- Responsible for the finances of the >100 person club, which organizes and funds weekly climbing events and multiple club outings per year.
- Implemented significant changes to the management of finances by implementing a standardized ledger, streamlining expenses, and enforcing monthly budgeting.
- Coordinated large gear purchase discounts with the Illinois Climbing association and other companies/associations.

2020-2021
Carle Hospital, University of Illinois at Urbana Champaign, Champaign, IL
Volunteer
- Assisted in the Student Ambassador Program, Oncology Department, and Post-Operations department, serving the greater local community of the University of Illinois.
- Interacted with patients, providing comfort, and provided general assistance to nursing staff with palliative care; restocked supplies and conducted various tasks as needed.

REFERENCES

Dr. David Koes
Department of Computational and Systems Biology
University of Pittsburgh / Carnegie Mellon University
E-mail: dkoes@pitt.edu

Dan DeBlasio
Department of Computational Biology
Carnegie Mellon University
E-mail: deblasio@cmu.edu

Dr. Emad Tajkhorshid
Department of Biochemistry
University of Illinois at Urbana-Champaign
E-mail: emad@illinois.edu